

bicycle

car

dog

S&DS 365 / 665
Intermediate Machine Learning

Course Wrap-up

April 22

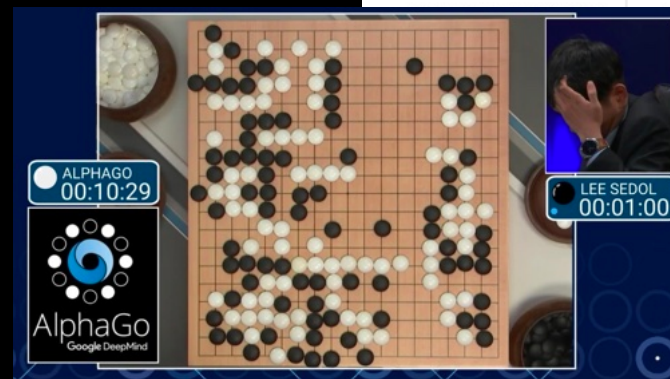
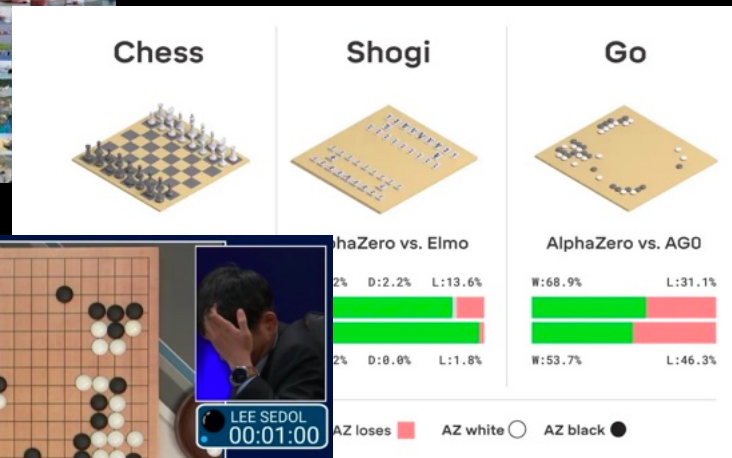
Yale

For today

- New directions in ML/AI
- Differing views of our AI future
- Final exam logistics
- IML Survivor

But first: The usual reminders

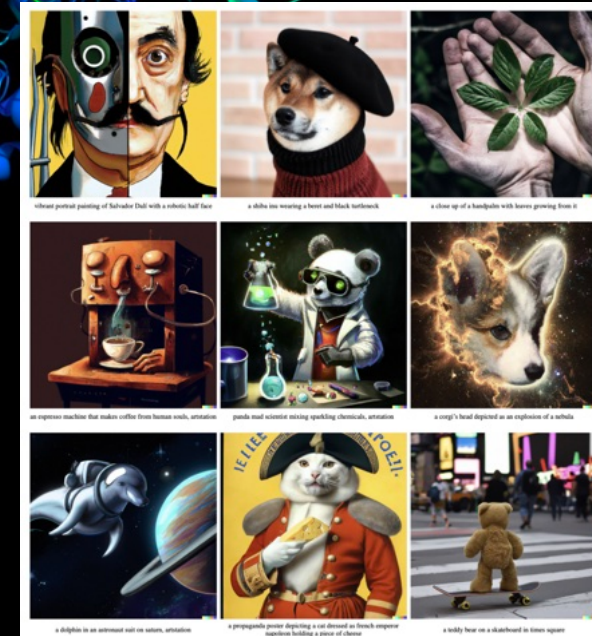
- Assn 5 due tomorrow
- Final exam: Tuesday, May 5, 2pm
- Practice exams posted



nature



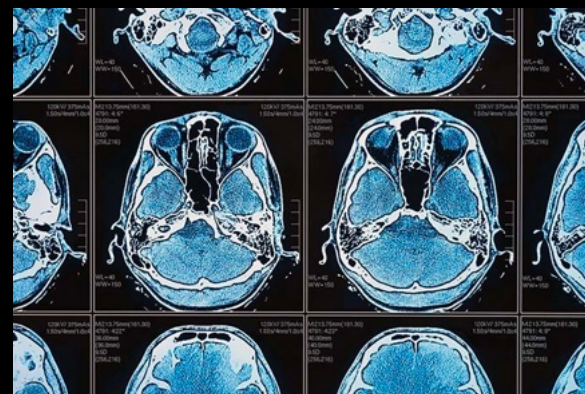
DeepMind



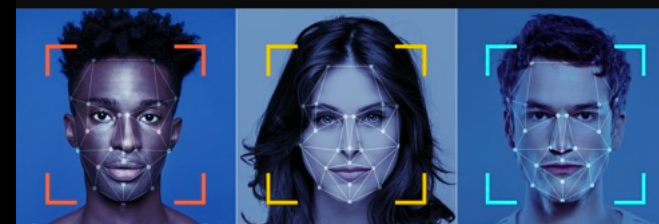
Ever growing reliance on ML



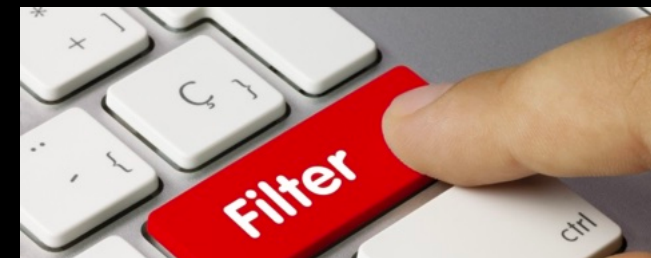
Self-driving Cars



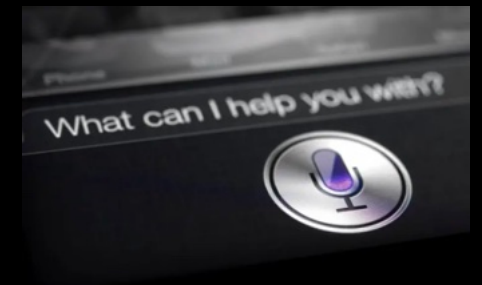
Medical Imaging



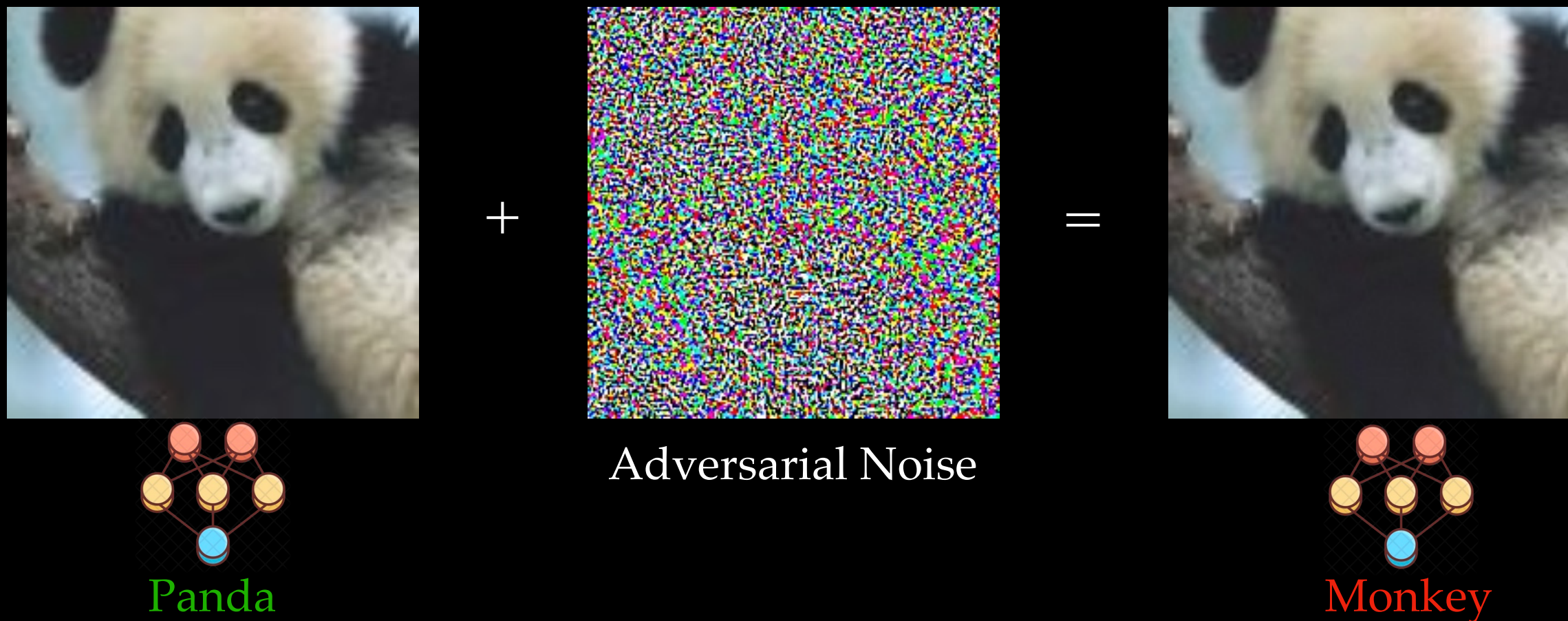
Face Recognition



Content Filtering



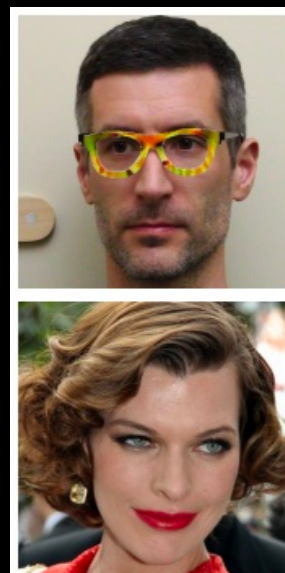
Speech Recognition



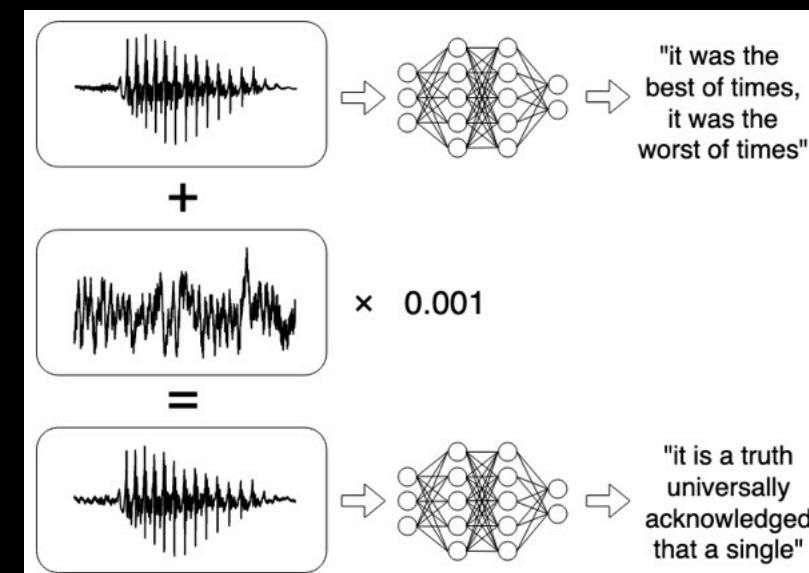
Bruna, Szegedy, Sutskever, Goodfellow, Zaremba, Fergus, Erhan (2013)
 Biggio, Corona, Maiorca, Nelson, Srndic, Laskov, Giacinto, Roli (2013)
 Goodfellow, Shlens, Szegedy (2015)



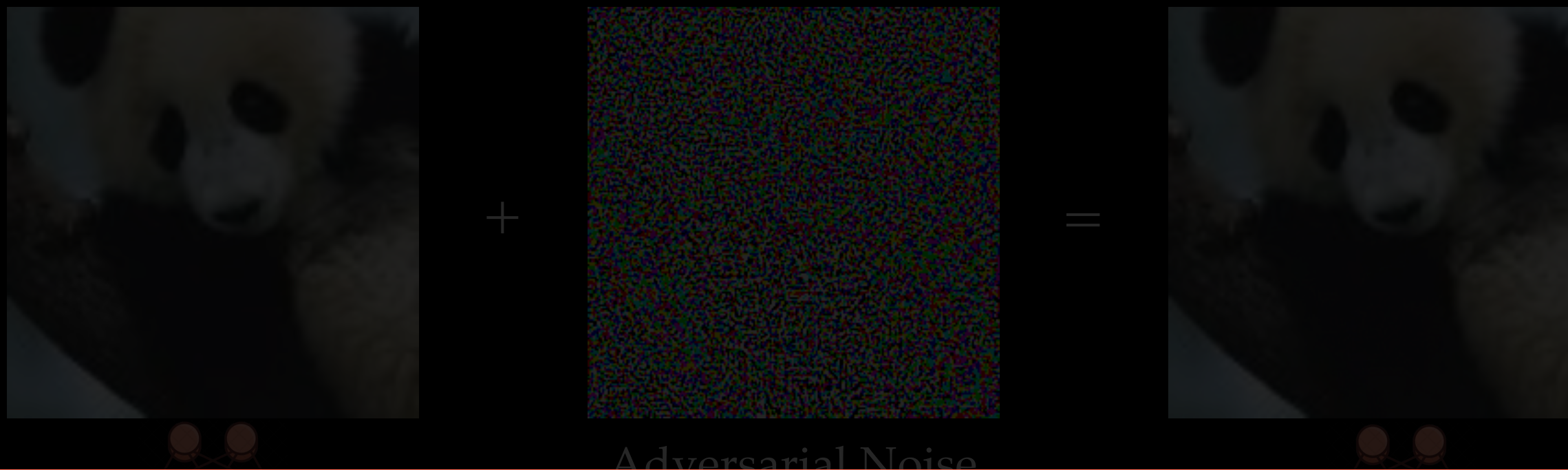
Eykholt, Evtimov, Fernandes, Li,
 Rahmati, Xiao, Prakash, Kohno, Song (2018)



Sharif, Bhagavatula, Bauer, Reiter (2018)



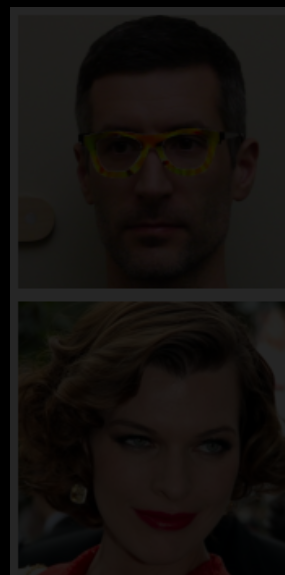
Carlini and Wagner (2018)



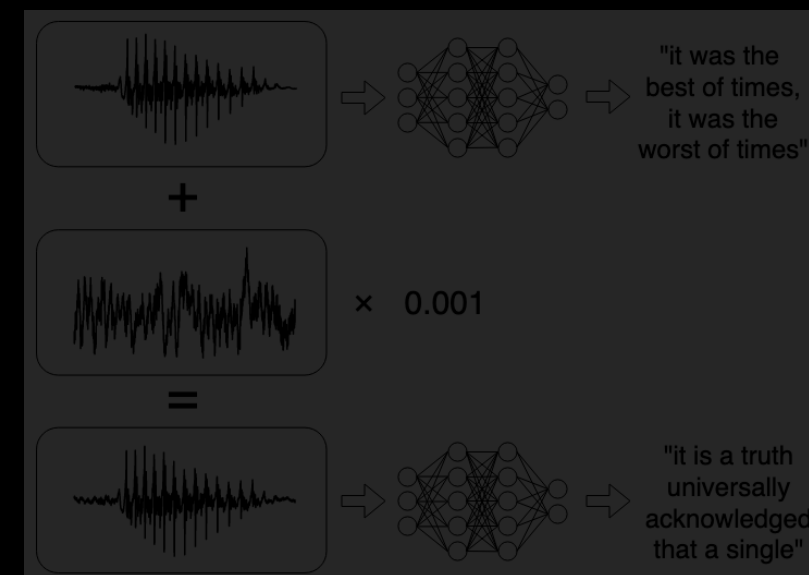
Can we learn predictors robust to adversarial examples? and how?



Eykholt, Evtimov, Fernandes, Li, Rahmati, Xiao, Prakash, Kohno, Song (2018)



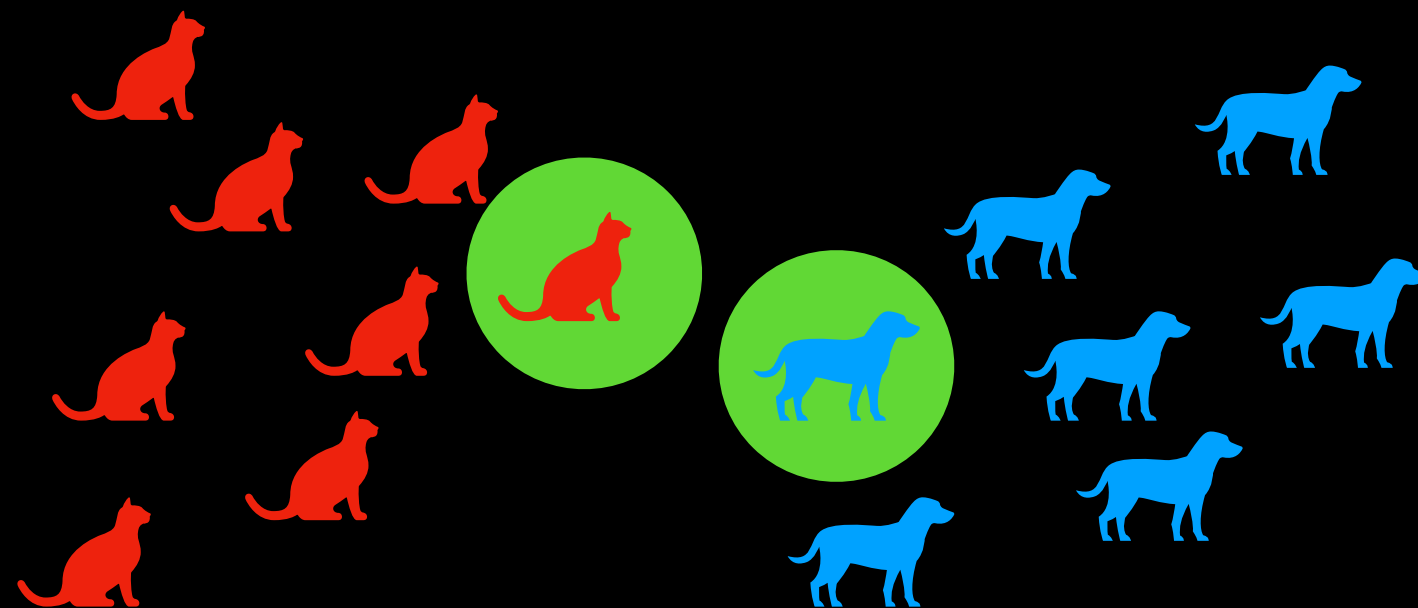
Sharif, Bhagavatula, Bauer, Reiter (2018)



Carlini and Wagner (2018)

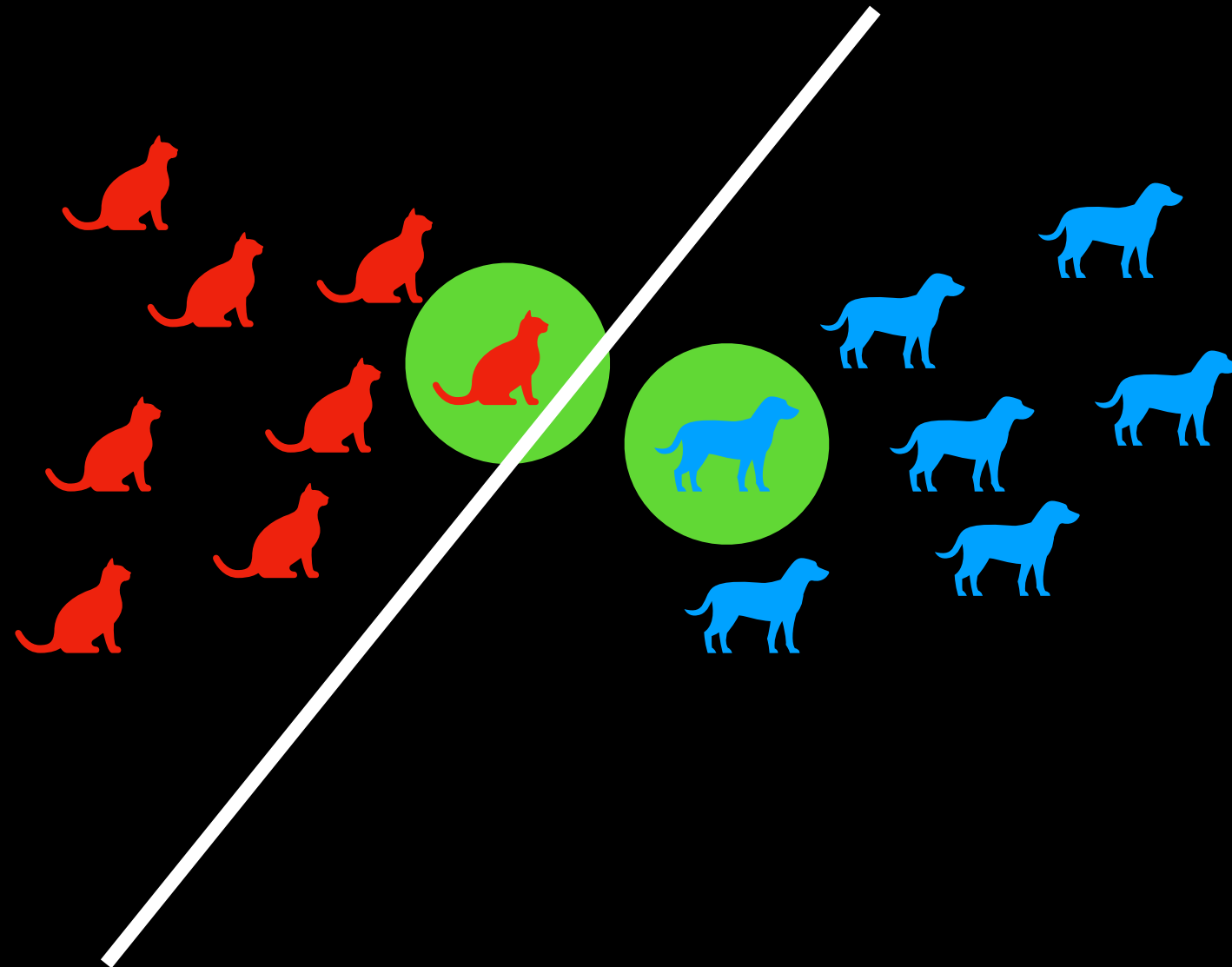
Adversarially Robust Learning

Using clean (non-corrupt) training examples to learn a predictor robust to adversarial attacks at test-time



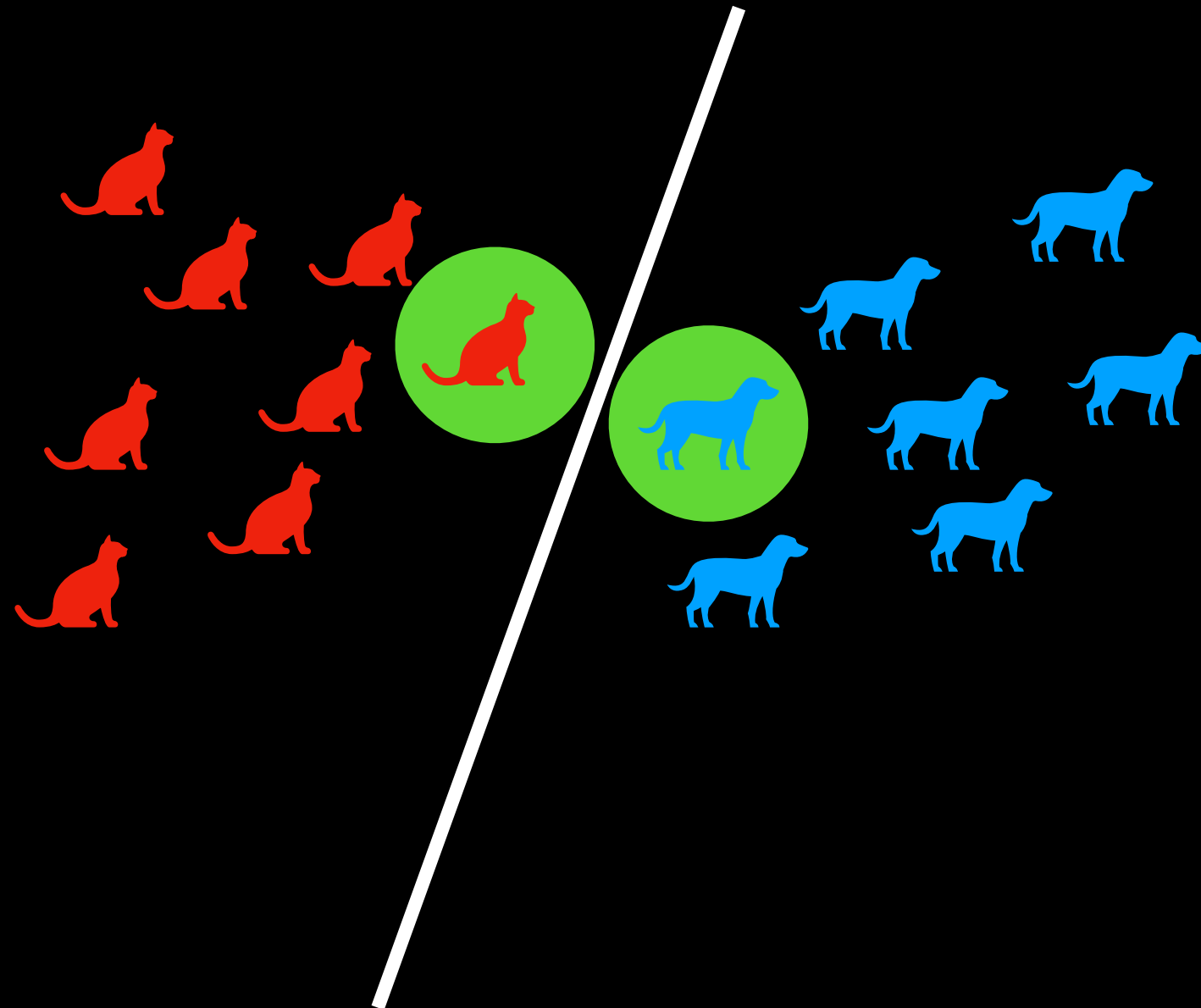
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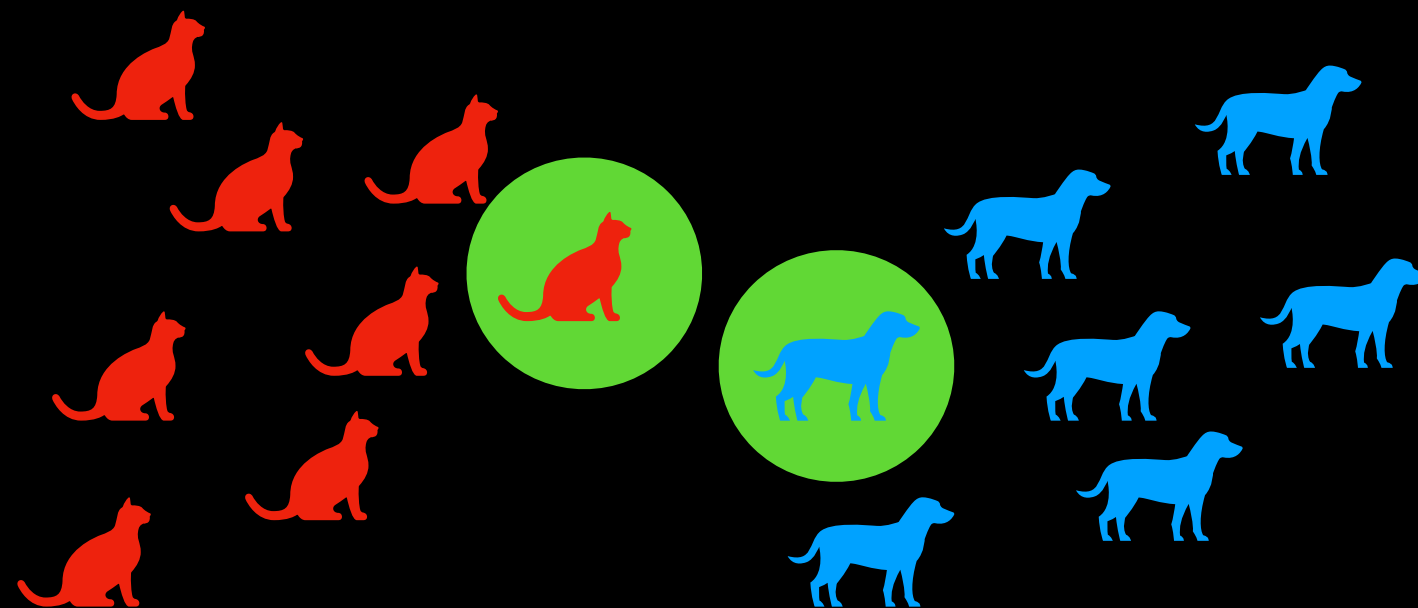
Adversarially Robust Learning

Using clean (non-corrupt) training examples to learn a predictor robust to adversarial attacks at test-time



Adversarially Robust Learning

Using clean (non-corrupt) training examples to learn a predictor robust to adversarial attacks at test-time



$$R_U(\hat{h}) = \mathbb{E}_{(x,y) \sim D} \left[\sup_{z \in U(x)} \mathbf{1}\{\hat{h}(z) \neq y\} \right]$$

Adversarially Robust Learning

Using iid clean (non-corrupt) training examples to learn a predictor robust to adversarial attacks at test-time

$$R_U(\hat{h}) = \mathbb{E}_{(x,y) \sim D} \left[\sup_{z \in U(x)} \mathbf{1}\{\hat{h}(z) \neq y\} \right]$$

Adversarial
Training

Optimize model parameters to fit training examples and their adversarial perturbations.

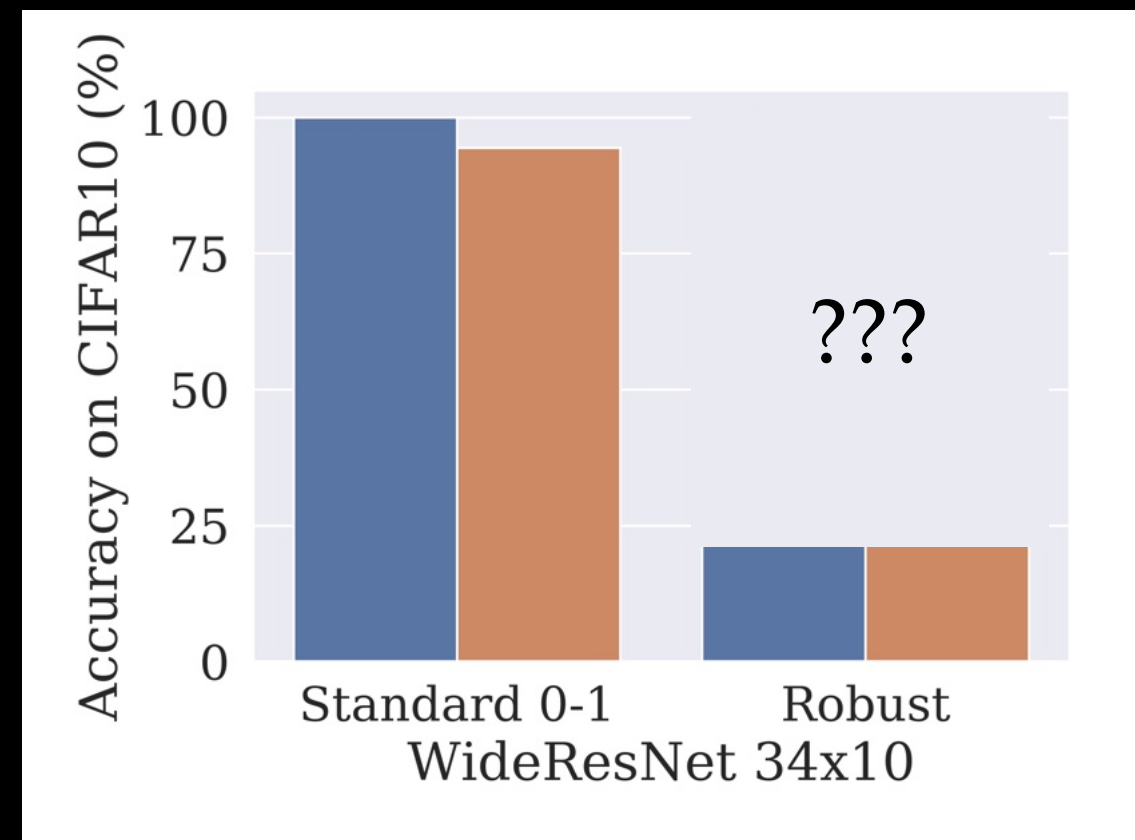
(Madry, Makelov, Schmidt, Tsipras, Vladu, 2017)
(Zhang, Yu, Jiao, Xing, El Ghaoui, Jordan, 2019)

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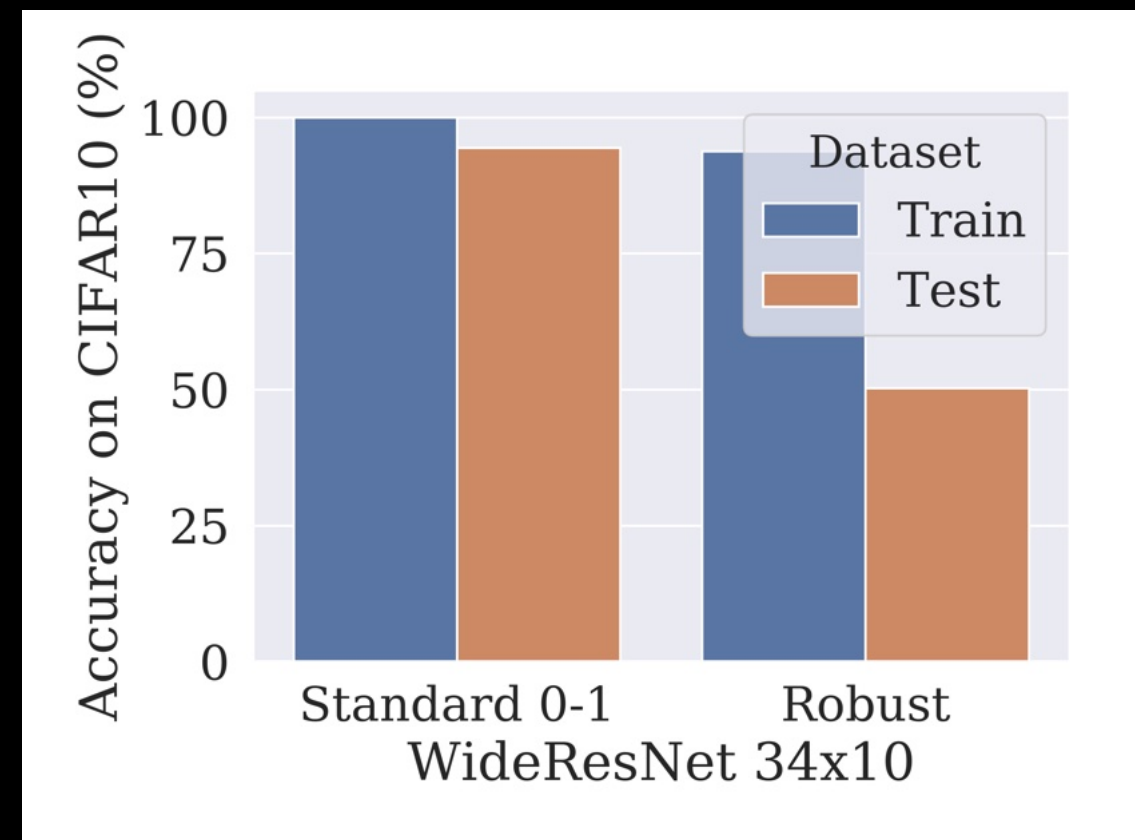
Schmidt, Santurkar, Tsipras, Talwar, and Madry (2018)

Adversarially Robust Learning

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Adversarially Robust Learning

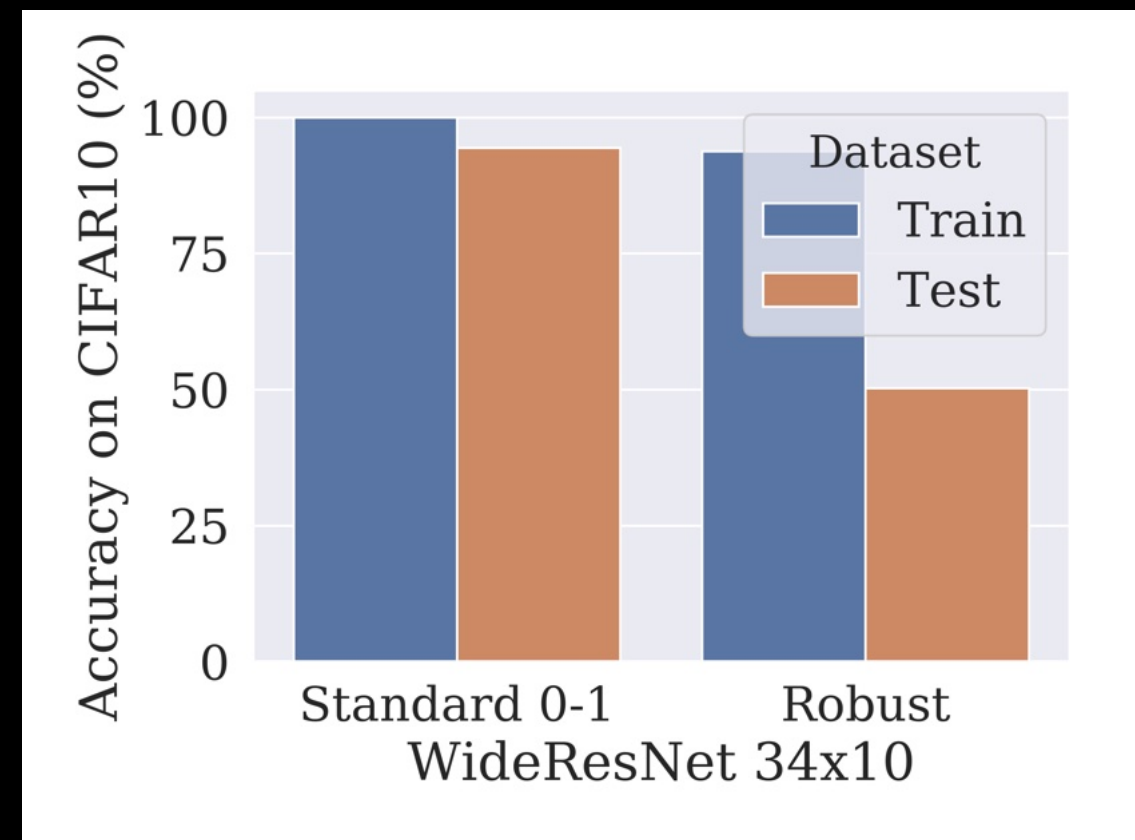
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M, Hanneke, Srebro (2019)

We provably show that *adversarial training* can **fail** to learn robust models.



Schmidt, Santurkar, Tsipras, Talwar, and Madry (2018)

Adversarially Robust Learning

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Optimize model parameters to fit training examples and their adversarial perturbations.

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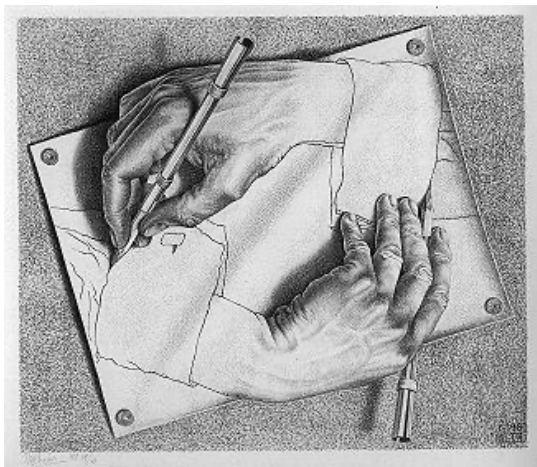
We provably show that *adversarial training* can **fail** to learn robust models.

Training an ensemble of models provably gives much stronger robustness guarantees.



Frontiers of ML and AI

For today's discussion, recall the view that ML is the technological core that underpins AI systems. An analogy would be the distinction between nuclear physics and nuclear energy.

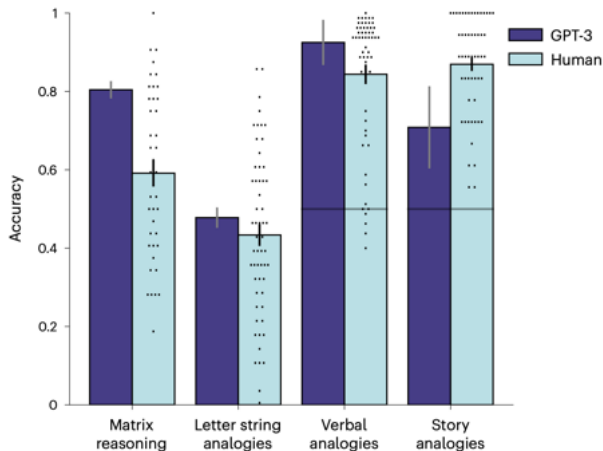


You scratch my back,
and I'll scratch yours

$$[A, B] = AB - BA = 0$$



LLMs have emergent abstraction abilities

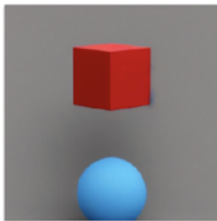


But LLMs can fail catastrophically

“Two shapes are in a room. The shape next to the cube is blue. The shape next to the sphere is red. Which is red, the cube or the sphere?”

But LLMs can fail catastrophically

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But LLMs can fail catastrophically

-  Based on the given information, it can be inferred that the sphere is red. The statement specifies that the shape next to the sphere is red, implying that the sphere itself is red. The color of the cube is not mentioned, so we cannot determine its color from the given information.

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Shortcomings are masked

- Innovations with Finetuning/RLHF hide these deficiencies
- System is trained to convince us
- Can lead to over-confidence and incorrect reasoning
- A central issue with “AI safety”

Another failure of relational reasoning

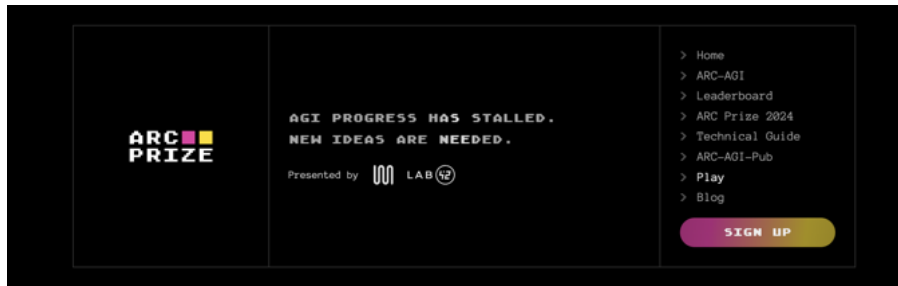


Two types of intelligence

- ① “Sensory”— acquire semantic and procedural knowledge
- ② “Relational”— rapid abstraction, association and generalization

*Can both types be supported in a single architecture?
This is a frontier question for AI*

ARC-AGI Prize

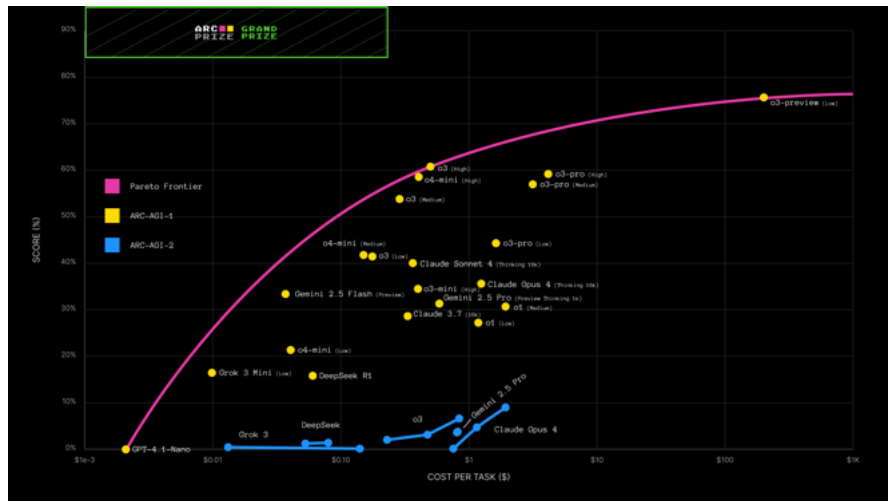


ARC-AGI Prize

The screenshot displays the ARC-AGI Prize interface, which is used for solving abstract reasoning tasks. It is divided into three main sections:

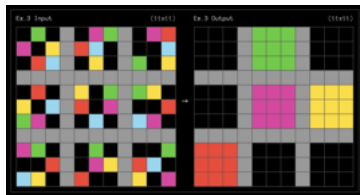
- Task demonstration:** This section on the left shows a 6x6 grid of 36 smaller 5x5 grids. Each of these smaller grids contains a unique pattern of colored squares (blue, red, green) on a black background, representing different examples of a task.
- Test input grid 1/1:** This section in the middle shows a single 5x5 grid with a pattern of yellow and black squares. It includes a "Next test input" button.
- Task controls:** This section on the right contains several controls:
 - Load task JSON:** Buttons for "Browse..." and "Random..." to load a specific task.
 - Task name:** Displays the current task name as "29623171.json" and the progress as "58 out of 400".
 - Show symbol numbers:** A checkbox that is currently unchecked.
 - Change grid size:** A dropdown menu set to "11x11" and a "Resize" button.
 - Action buttons:** "Copy from input", "Reset grid", and a green "Submit!" button.
 - Output grid:** A large, empty 11x11 grid at the bottom right where the solution is displayed.

ARC-AGI Prize



“Platonic” abstractions

- What is required to discover *useful* “cognitive wheels”?
- Platonic view of primitive abstractions (small natural numbers, spatial relations) — they are real and exist independent of human thought



- Algorithms should be encouraged (through inductive biases?) to employ these abstractions

Chollet, “On the measure of intelligence” (2019); Chollet, “OpenAI o3 Breakthrough High Score on ARC-AGI-Pub, 2024. <https://arcprize.org/blog/oai-o3-pub-breakthrough>, Beger et al. (2025) “Do AI models perform human-like abstract reasoning across modalities?” arXiv:2510.02125

“Platonic” abstractions

“Using such abstractions is how we humans are able to understand our world, predict probable outcomes of our decisions, and to efficiently deal with novelty. These are all things we want AI systems to do in more trustworthy ways.”

Creativity and rigorous reasoning

- We need new ML ideas to enable abstraction and analogy
- Humans often resist logical and evidence-based reasoning
- Can AI be based on more “rigorous” reasoning?

Proofs with AI

Early science acceleration experiments with GPT-5

Sébastien Bubeck¹, Christian Coester², Ronen Eldan¹, Timothy Gowers³, Yin Tat Lee¹,
Alexandru Lupascu^{1,4}, Mehtaab Sawhney⁵, Robert Scherrer⁴, Mark Sellke^{1,6},
Brian K. Spears⁷, Derya Unutmaz⁸, Kevin Wei¹, Steven Yin¹, Nikita Zhivotovskiy⁹

¹OpenAI

²University of Oxford

³Collège de France and University of Cambridge

⁴Vanderbilt University

⁵Columbia University

⁶Harvard University

⁷Lawrence Livermore National Laboratory

⁸The Jackson Laboratory

⁹University of California, Berkeley

November 20, 2025

SHORT PROOFS IN COMBINATORICS AND NUMBER THEORY

BORIS ALEXEEV, MOE PUTTERMAN, MEHTAAB SAWHNEY, MARK SELLKE, AND GREGORY VALIANT
OPENAI

ABSTRACT. We give a triplet of short proofs, each of which answers a question raised by Erdős. The first concerns the small prime factors of $\binom{n}{k}$, the second concerns whether an additive basis A can always be split into pieces A_1 and A_2 such that each of $A_1 + A_1$ has bounded gaps, and the final concerns whether $\{np\}$ is “well-distributed” in the sense introduced by Hlawka and Petersen. In each case, the proof is due entirely to an internal model at OpenAI.

SHORT PROOFS IN COMBINATORICS, PROBABILITY AND NUMBER THEORY II

BORIS ALEXEEV, MOE PUTTERMAN, MEHTAAB SAWHNEY, MARK SELLKE, AND GREGORY VALIANT
OPENAI

ABSTRACT. We give a quintet of proofs resulting from questions posed by Erdős. These questions concern ordinary lines in phase point sets, sequences with uniformly small exponential sums, K_4 -free 4-critical graphs with few chords in any cycle, a counterexample to a “fewnomial” version of the Erdős-Turán discrepancy bound, and a finiteness theorem for integers n such that $n - nk^2$ is prime for all $k \leq \sqrt{n/a}$ coprime to n (for fixed $a \in \mathbb{Z}_+$). Each proof is due to an internal model at OpenAI.

Proofs with AI

Run a sub-Markov chain on the divisibility poset. Primitive (antichain) $\Rightarrow$ each path hits A at most once. $\Rightarrow$



Primitive sets above x

Main Theorem. If $A \in \mathcal{N} \cap [x, \infty)$ is primitive, then

$$\sum_{a \in A} \frac{1}{a \log a} \leq 1 + O\left(\frac{1}{\log x}\right),$$

so in particular $\sum_{a \in A} \frac{1}{a \log a} < 1 + o(1)$



① Key arithmetic input

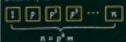
von Mangoldt function:
 $\Lambda(q) = \log p$ if $q = p^k$
 and 0 otherwise.

$$\sum_{q|n} \Lambda(q) = \log n$$

Lemma (estimates we need)

$$\begin{aligned}\psi(x) &= \sum_{n \leq x} \Lambda(n) \ll x \\ A(t) &= \sum_{q \leq t} \frac{\Lambda(q)}{q} = \log t + O(1) \\ T(m, y) &= \sum_{q \leq y} \frac{\Lambda(q)}{q \log^2(mq)} \\ &= \frac{1}{\log(my)} + O\left(\frac{1}{\log^2(my)}\right)\end{aligned}$$

Divisors of n .



② Build a sub-Markov chain
on the divisibility poset

Choose Y large enough. For $m \geq X$ and $q \geq Y$,

$$p(m, m_q) = \frac{\log m \cdot \Lambda(q)}{q \log^2(m_q)}$$

$$\sum_i p(m, m_i) \leq 1 \quad (\text{sub-Markov})$$



Every jump is strict, so visited states form a divisibility chain

$$n_0 \mid n_1 \mid n_2 \mid \dots$$

③ Inject source mass

$$b_x(n) = \frac{1}{n \log n} \left[\underbrace{\sum_{\substack{q|n \\ q < Y}} \Lambda(q)}_{\text{small-prime}} + \underbrace{\sum_{\substack{q|n \\ q > Y, \frac{n}{q} < x}} \Lambda(q)}_{\text{first-entry}} \right]$$

$$\triangleright B_x = \sum_{n \leq x} b_n(n) = 1 + O\left(\frac{1}{\log x}\right)$$

$$\mu_x(n) = \frac{b_x(n)}{\beta_x}$$

Initial / source mass



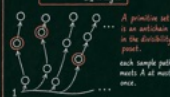
④ Hitting probabilities and the primitive-set argument

Define $v_x(n) = P(\text{the chain ever visits } n)$.
Then for $n \geq x$,

$$v_n(n) = \mu_n(n) + \sum_{\substack{q|n \\ q \geq 2, n/q \geq 1}} v_n(n/q) p(n/q, n).$$

Induction on n (by size) gives

$$v_x(n) = \frac{1}{B_x n \log n}$$



$$\sum_{a \in A} v_x(a) \leq 1$$

$$\Rightarrow \sum_{a \in A} \frac{1}{a \log a} = B_x \sum_{a \in A} v_x(a) \leq B_x$$

$$\Rightarrow \sum_{a \in A} \frac{1}{a \log a} \leq 1 + O\left(\frac{1}{\log x}\right)$$

Proofs with AI

First Proof

A set of ten math questions to evaluate the capabilities of AI systems to autonomously solve problems that arise naturally in the research process.

About the Project

In baking, the *first proof*, or bulk fermentation process, is a crucial step in which one lets the entire batch of dough ferment as one mass, before dividing and shaping it into loaves.

This project represents our preliminary efforts to develop an objective and realistic methodology for assessing the capabilities of AI systems to autonomously solve research-level math questions. After letting these ideas ferment in the community, we hoped to produce a more structured benchmark.

We presented a diverse set of 10 research-level math questions, drawn from algebraic combinatorics, spectral graph theory, algebraic topology, stochastic analysis, symplectic geometry, representation theory, lattices in Lie groups, tensor analysis, and numerical linear algebra. Each question arose naturally in the research process of the authors and has been answered with a proof of roughly five pages or less.

AI and humanity: What do *you* think?



- ① AI is just a new technology, like electricity, and civilization will advance much like it has for the last several hundred years; nothing will fundamentally change.
- ② The world will fundamentally change, but humanity will find ways to safely control supernatural intelligent agents; AI will safely and peacefully improve the human condition.
- ③ AI poses a significant existential threat; humanity must learn to co-exist with supernatural intelligent agents or be wiped out by them.
- ④ Something completely different from the above three.

AI and you



- 1 AI will complement my future job, but I don't expect it to significantly change what I do in the future.
- 2 I need to reconsider my career goals and dreams, because AI is likely to dramatically change my planned career

We've covered a lot of ground!

Week	Dates	Topics	Demos & Tutorials	Lecture Slides	Readings & Notes	Assignments & Exams
1	Jan 12, 14	Sparse regression	 Python elements  Pandas and regression  Lasso example	Mon: Course overview Wed: Sparse regression	Google Colab Basics PML Section 11.4 Notes on linear regression	
2	Jan 21, 23	Smoothing, kernels, and Density estimation	 Smoothing example  Using different kernels  Density estimation demo	Wed: Lasso (continued) and smoothing Fri: Smoothing and density estimation	PML Sections 16.3, 17.1 Notes on computing the lasso Risk bounds for local smoothing	Quiz 1
3	Jan 26, 28	Mercer kernels and Neural networks	 Mercer kernels (1/3)  Mercer kernels (2/3)  Mercer kernels (3/3) TensorFlow playground	Mon: Mercer kernels Wed: Kernels and Neural networks	Notes on Mercer kernels Notes on backpropagation	 Assn 1  ipynb converter

We've covered a lot of ground!

4	Feb 2, 4	Overparameterized models and Convolutional neural networks	 np-complete example (1/2)  np-complete example (2/2)  Convolution demo	Mon: Double descent Wed: Convolutional neural networks	PML Sections 13.1, 13.2 Notes on double descent	Quiz 2
5	Feb 9, 11	Gaussian processes	 CNN demo (1/2)  CNN demo (2/2)  Parametric Bayes  Gaussian processes	Mon: CNNs and Gaussian Processes Wed: Gaussian processes	PML Section 17.2 Notes on Bayesian inference Notes on nonparametric Bayes	Assn 1 in  Assn 2  ipynb converter
6	Feb 16, 18	Approximate inference and variational inference	 Gibbs sampling for image denoising	Mon: Recap of GPs Introduction to approximate inference Wed: Variational inference	Notes on simulation PML Section 20.3 Notes on variational inference	Quiz 3
7	Feb 23, 25	Variational autoencoders, Graphs and structure learning	 Variational autoencoders  Graphical lasso demo	Mon: VAEs Wed: Sparsity and graphs	Notes on graphs and structure learning	Assn 2 in  ipynb converter

We've covered a lot of ground!

8	Mar 2	Graphs and structure learning		Mon: Discrete data and graph neural nets	Graph neural networks PML Section 23.4	
8	Mar 4	Midterm	Wednesday in-class			Mar 4: Midterm CO Assn 3 out CO ipynb converter
9	Mar 9, 11	No class, Spring Break				
10	Mar 16, 18	No class, Spring Break				
11	Mar 23, 24	Deep reinforcement learning	CO Q-learning demo CO DQN demo	Mon: Reinforcement learning Wed: Deep reinforcement learning	Sutton and Barto, Section 6.5	Assn 3 in CO Assn 4 out CO ipynb converter
12	Mar 30, Apr 1	Policy methods	CO Policy gradients demo CO Actor-critic demo	Mon: Policy gradient methods Wed: Policy gradients (continued)	Sutton and Barto, Section 13.1-13.3, 13.5	Quiz 4

We've covered a lot of ground!

13	Apr 6, 8	Sequential models	 vanilla RNN  Fakespeare GRU	Mon: HMMs and RNNs Wed: RNNs, GRUs, LSTMs, and all that	TensorFlow: Text generation Notes on HMMs and Kalman filters PML Chapter 15	Assn 4 in  Assn 5 out  ipynb converter
14	Apr 13, 15	Sequence-to-sequence models and Transformers	 GPT-4 Python API	Mon: Sequence models and attention Wed: Transformers, LLM scaling PML Sections 15.4, 15.5		Quiz 5
15	Apr 20, 22	The LLM pipeline; broader issues	 Transformer demo Minimal LLM decoder	Mon: LLM finetuning, postprocessing Wed: Course wrap up		Assn 5 in
17	May 5, 2pm	Final exam				Registrar: final exam schedule

Final exam

- Length: About $1.5\times$ midterm
- Cumulative, but emphasis on material after midterm
- Closed book, cheat-sheet (2-sided, handwritten)





- Majority votes off a topic!
- What are you thinking? Persuade your classmates!

Vote a topic off the final!

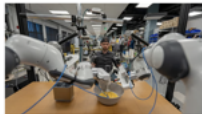


- 1 Gaussian processes
- 2 Double descent
- 3 Reproducing kernel Hilbert spaces
- 4 Variational autoencoders
- 5 Graphical lasso
- 6 Deep Q-Learning
- 7 Policy gradient methods
- 8 GRUs

Advancing ML: Something for everyone

- Designing new methods
- Applying to new domains
- Solving deep mathematical puzzles
- Tackling unique engineering challenges
- Designing interfaces
- Collecting data
- Equitable use through policy and law
- Outreach and communication to broad communities

Always news



Science & technology

Robots can learn new actions faster thanks to AI techniques

They could soon show their moves in settings from car factories to care homes



Finance & economics

Computers unleashed economic growth. Will artificial intelligence?

Two years after ChatGPT-3.5 arrived, progress has been slower than expected



Business

Nvidia's boss dismisses fears that AI has hit a wall

But it's "urgent" to get to the next level, Jensen Huang tells *The Economist*



Babbage

How AI will lead to a new scientific renaissance

Our podcast on science and technology. Nobel laureates Demis Hassabis and Jennifer Doudna, plus Google's James Manyika, present their vision for how AI is transforming science



Business

What ChatGPT's corporate victims have in common

The first casualties of generative AI offer lessons for other businesses



Science & technology in 2025

Supercomputers and AI are helping build better climate models

The coming year will be a big one for the advance of climate science



Science & technology in 2025

AI will boost drug development in 2025

Software-inspired medicines are getting closer to prime time



Finance in 2025

There will be no immediate productivity boost from AI

Everyone calm down a little

Your input

- Please complete a course review!
- I greatly value your comments and feedback
- Feel free to send me comments privately
- Let me know how you use and continue to learn ML!

Thank you!